

Canadian Model for Peatlands Bulk Density Modelling - 2024

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Header 1

Header 2

```
## Warning: attribute variables are assumed to be spatially constant throughout
## all geometries
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```

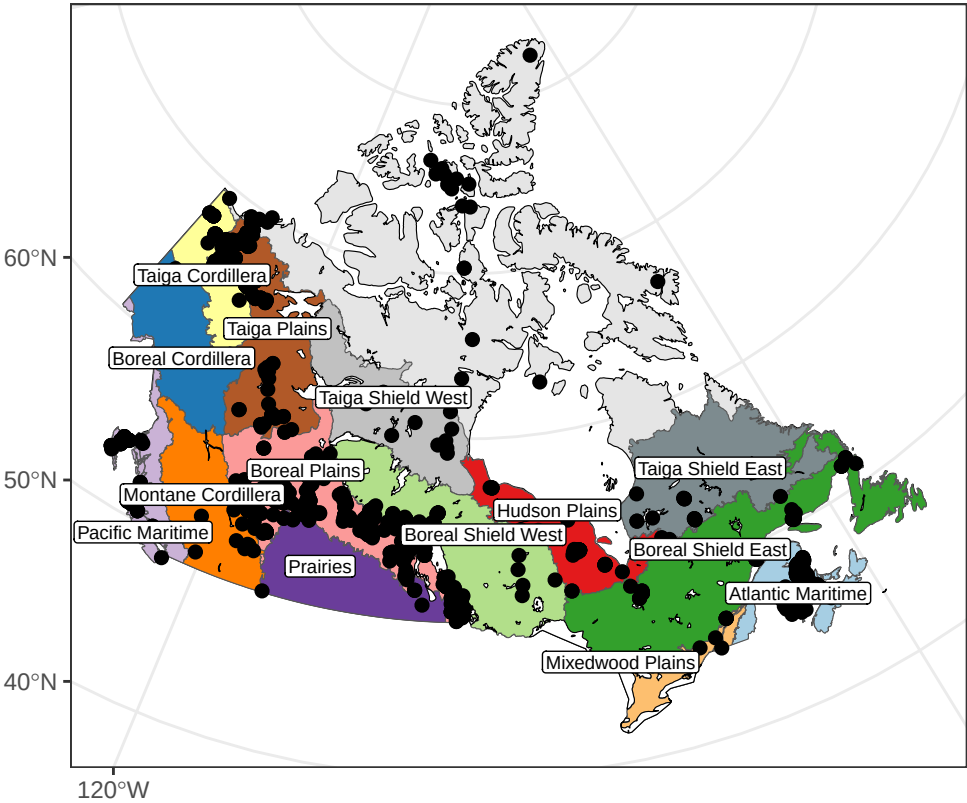


Figure 1: Locations of NDVI time series data plots.

```
## # A tibble: 36 x 4
```

```

## # Groups:   ECOZONE_NAME_EN [15]
##   ECOZONE_NAME_EN  CaMPNutrient_2024  ORG_DEPTH      n
##   <chr>           <fct>                <dbl> <int>
## 1 ArcticCordillera permafrost             50      1
## 2 AtlanticMaritime bog                   450     231
## 3 AtlanticMaritime rich fen               290      15
## 4 AtlanticMaritime swamp                 240       1
## 5 BorealCordillera permafrost             80       1
## 6 BorealPlains     bog                   200.     58
## 7 BorealPlains     permafrost            226      49
## 8 BorealPlains     poor fen               247      25
## 9 BorealPlains     rich fen               204     191
## 10 BorealPlains     swamp                 173      14
## # i 26 more rows

## # A tibble: 129 x 6
## # Groups:   ECOZONE_NAME_EN, CaMPNutrient_2024, TREED [77]
##   ECOZONE_NAME_EN  CaMPNutrient_2024  TREED PROV_TERR  ORG_DEPTH      n
##   <chr>           <fct>                <chr> <chr>         <dbl> <int>
## 1 ArcticCordillera permafrost          U    NU             50       1
## 2 AtlanticMaritime bog                  N    NB            450     219
## 3 AtlanticMaritime bog                  N    NS            850       1
## 4 AtlanticMaritime bog                  U    NB            250       3
## 5 AtlanticMaritime bog                  Y    NB            380       8
## 6 AtlanticMaritime rich fen             N    NB            285      14
## 7 AtlanticMaritime rich fen             Y    NB            380       1
## 8 AtlanticMaritime swamp                N    NB            240       1
## 9 BorealCordillera permafrost           Y    YT             80       1
## 10 BorealPlains     bog                  N    AB           232.       4
## # i 119 more rows

## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula:
## PeatC_dens ~ log(DEPTHCLASS) + TREED + (1 | CORE_ID) + CaMPNutrient_2024 +
##   ECOZONE_NAME_EN
## Data: PROFILE_2024_w_site_info
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## -198833.5 -198629.4  99440.8 -198881.5    36514
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.8020 -0.5407 -0.0489  0.4921 26.4030
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   CORE_ID  (Intercept) 0.0001076 0.01038
##   Residual                0.0002378 0.01542
## Number of obs: 36538, groups:  CORE_ID, 1194
##
## Fixed effects:
##
##              Estimate Std. Error t value
## (Intercept)    0.0363715  0.0137943   2.637
## log(DEPTHCLASS) 0.0063398  0.0001005  63.085

```

```

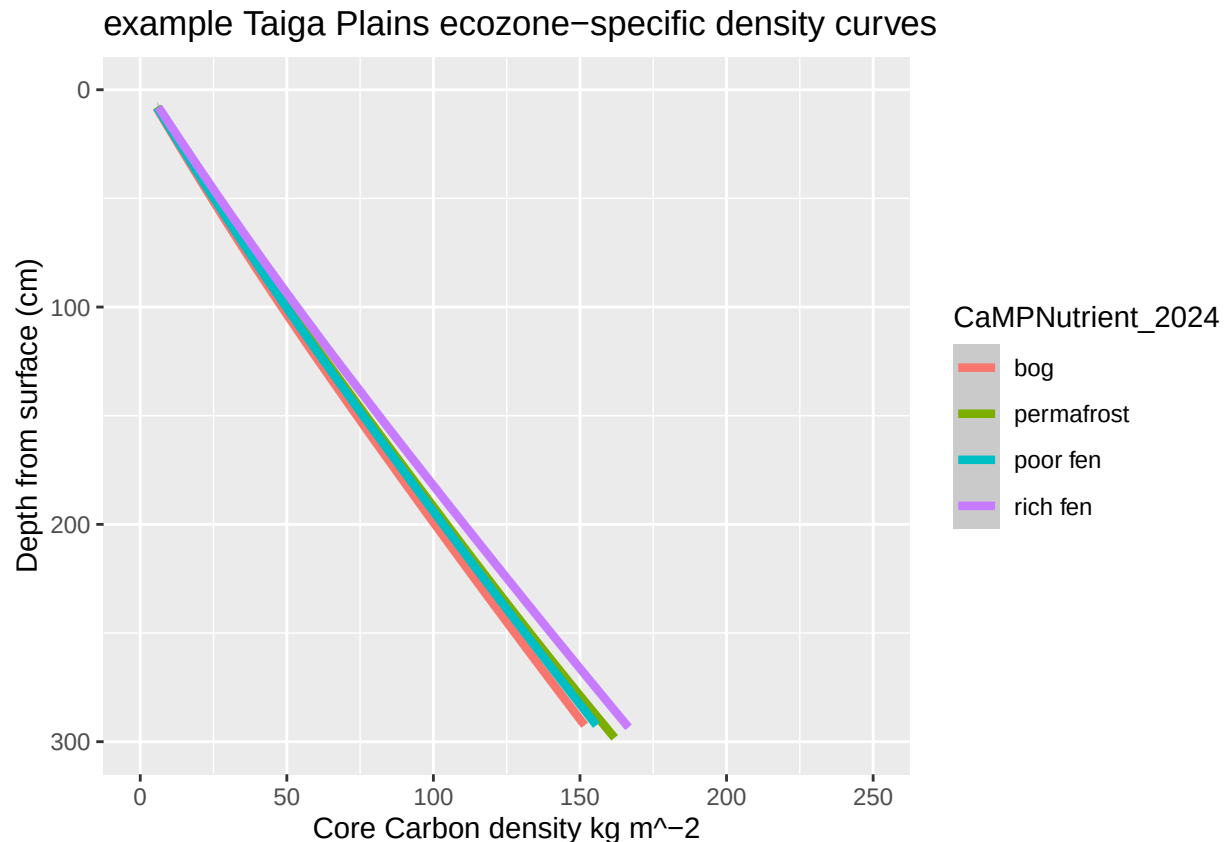
## TREEDU -0.0004113 0.0012079 -0.340
## TREEDY 0.0019315 0.0009037 2.137
## CaMPNutrient_2024permafrost 0.0014830 0.0013218 1.122
## CaMPNutrient_2024poor fen 0.0013367 0.0017535 0.762
## CaMPNutrient_2024rich fen 0.0049942 0.0010675 4.678
## CaMPNutrient_2024swamp 0.0095246 0.0025007 3.809
## ECOZONE_NAME_ENAtlanticMaritime -0.0074366 0.0138094 -0.539
## ECOZONE_NAME_ENBorealCordillera -0.0122177 0.0185384 -0.659
## ECOZONE_NAME_ENBorealPlains -0.0005050 0.0137785 -0.037
## ECOZONE_NAME_ENBorealShield 0.0002046 0.0137604 0.015
## ECOZONE_NAME_ENHudsonPlains -0.0026670 0.0138098 -0.193
## ECOZONE_NAME_ENMixedwoodPlains 0.0082958 0.0142213 0.583
## ECOZONE_NAME_ENMontaneCordillera -0.0009663 0.0142714 -0.068
## ECOZONE_NAME_ENNorthernArctic 0.0090807 0.0141160 0.643
## ECOZONE_NAME_ENPacificMaritime 0.0103851 0.0138183 0.752
## ECOZONE_NAME_ENPrairies -0.0009621 0.0146692 -0.066
## ECOZONE_NAME_ENSouthernArctic 0.0016024 0.0139278 0.115
## ECOZONE_NAME_ENTaigaCordillera -0.0030161 0.0139070 -0.217
## ECOZONE_NAME_ENTaigaPlains 0.0007347 0.0137540 0.053
## ECOZONE_NAME_ENTaigaShield 0.0070282 0.0138893 0.506

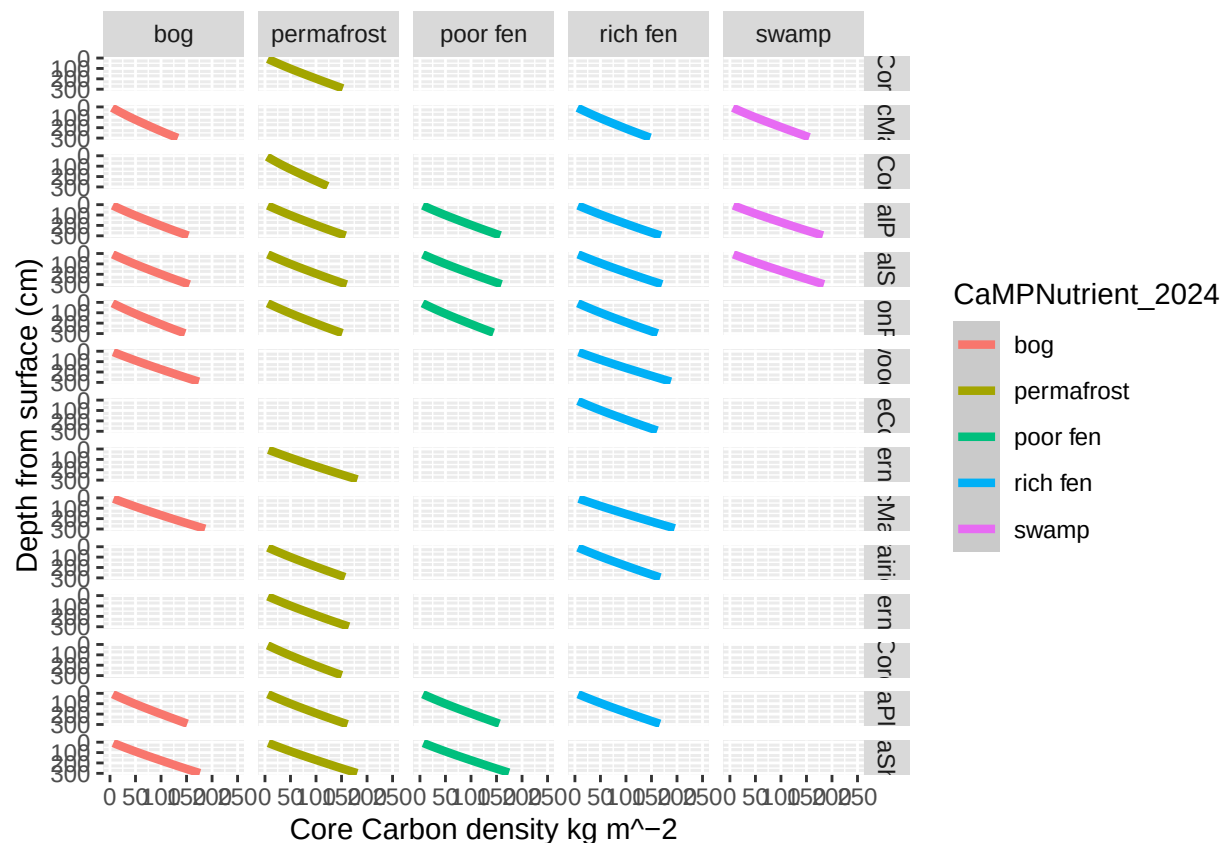
```

```

##
## Correlation matrix not shown by default, as p = 22 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)         if you need it

```





```
## # A tibble: 72 x 3
##   UniquePeatCombo      term estimate
##   <chr>              <chr>    <dbl>
## 1 ArcticCordillera-permafrost a      0.378
## 2 ArcticCordillera-permafrost b      1.06
## 3 AtlanticMaritime-bog      a      0.281
## 4 AtlanticMaritime-bog      b      1.08
## 5 AtlanticMaritime-rich fen a      0.347
## 6 AtlanticMaritime-rich fen b      1.06
## 7 AtlanticMaritime-swamp    a      0.391
## 8 AtlanticMaritime-swamp    b      1.05
## 9 BorealCordillera-permafrost a      0.259
## 10 BorealCordillera-permafrost b      1.08
## # i 62 more rows
```

Below are two code chunks used for gap-filling bulk density in the Bauer et al 2024 national peat core composite:

And an archive of the bulk density modelling from Bona et al 2020: